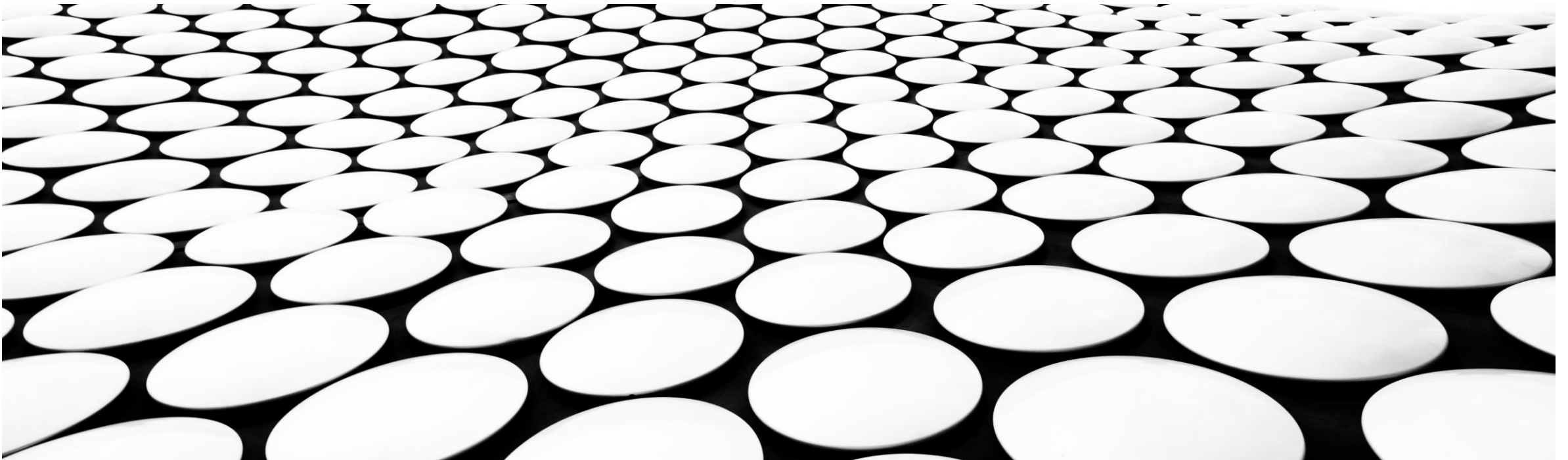


IOT DESIGN AND PROTOTYPING (420-N55-LA)

OMID PANAHI UPDATED DOCUMENT





INTRODUCTION TO THE COURSE



COURSE INFORMATION

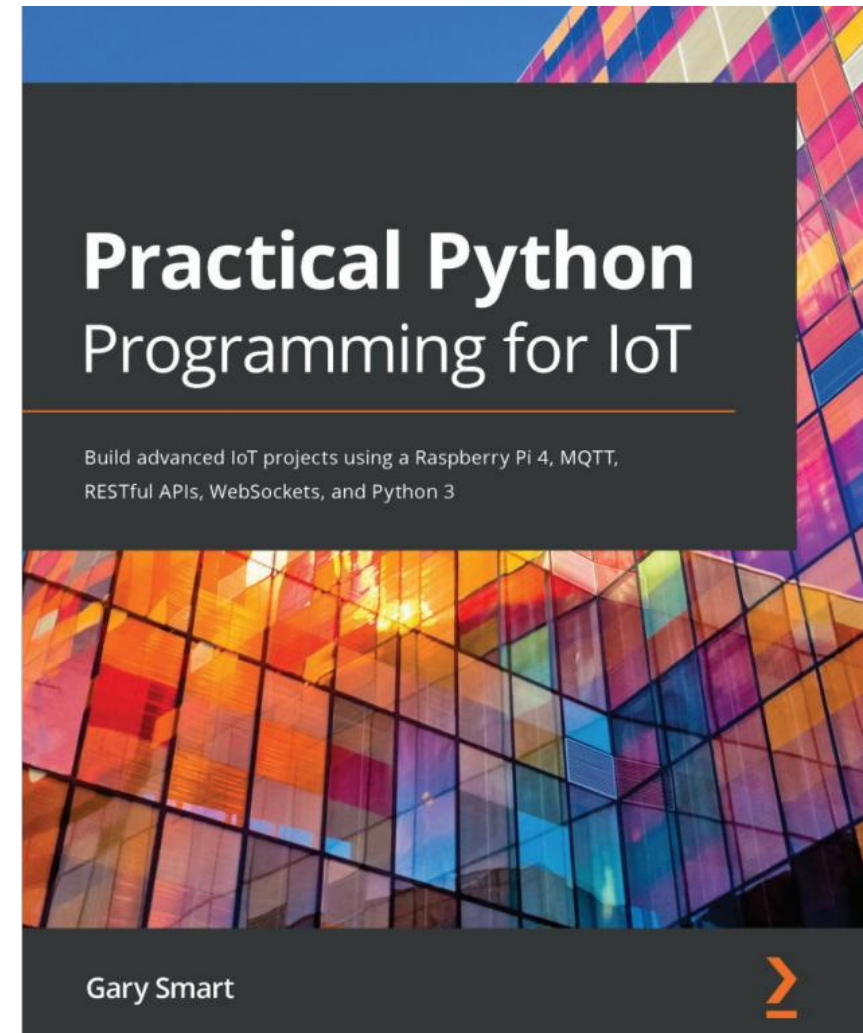
Course title	IoT: Design and Prototyping of Connected Devices		
Course number	420-N55-LA		
Credits:	1.33		
Hours:	45		
Weighting:	1-2-1		
Modality:	Classroom and laboratory / In-Person		
Prerequisites:	420-N35-LA Introduction to IoT (Internet of Things) and Embedded Systems	Co-requisites:	

OVERVIEW

- This course explores the design and creation of interconnected digital ecosystems. Starting with a solid understanding in n-tier application design, students will be introduced to Raspberry-Pi, the adaptable Raspbian OS, Docker, MySQL, and MongoDB. Beginning with essential Python basics, the curriculum advances to detailed studies of Python paradigms specifically crafted for IoT.
- Participants will navigate the complexities of the Raspberry Pi platform, unlocking its potential with Python-driven GPIO operations.
- They will also grasp the essence of web services, utilizing Python as both a client and a service, and master RESTful web service connections as well as Spring-based implementations.
- This journey enhances students' understanding of databases, emphasizing the integration of MySQL with Python for optimal data storage. As students stride towards a secure digital future, they'll be equipped to fortify their IoT projects against potential threats.
- The course culminates in hands-on project presentations, celebrating their foray into the IoT realm.

COURSE MATERIAL

- Textbook
- Practical Python Programming for IOT by Gary Smart. Packt Publishing 2020. ISBN 978-1-83898-246-1



REQUIRED HARDWARE

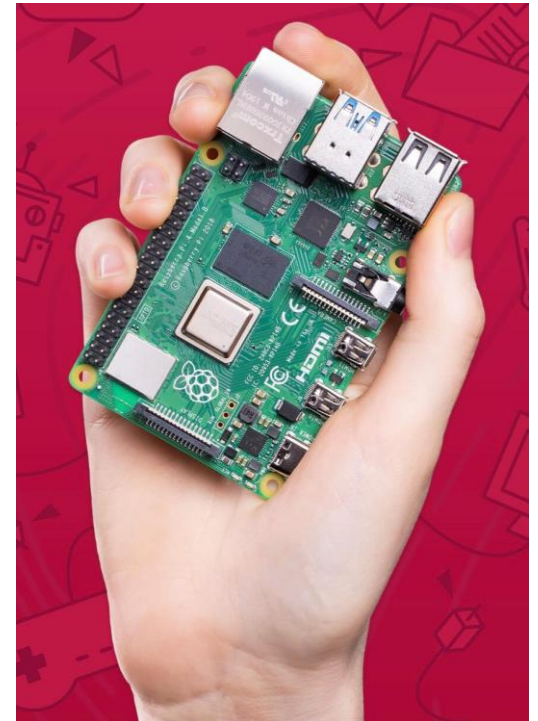
1. PC/LAPTOP

- You need this for basic development much like your other courses.
- Mac or PC, whatever.

2. Raspberry Pi 5 (no other version is accepted)

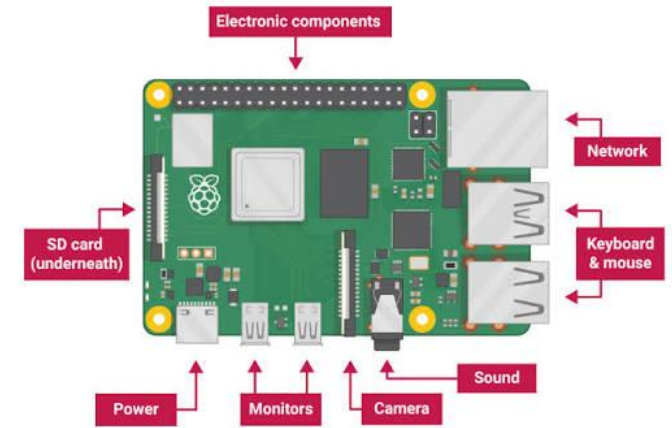
1. Mandatory

- Raspberry Pi 5 board
- Memory Card (32GB minimum)
- Card Reader
- Power Supply (45W)



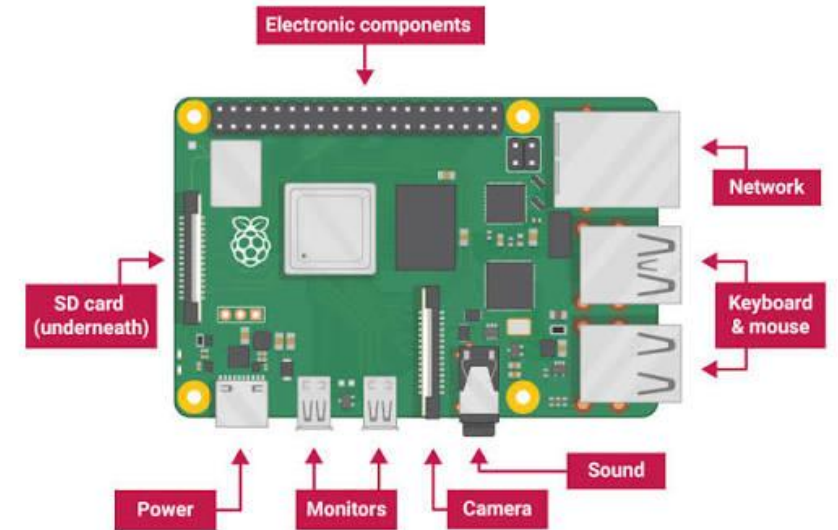
RASPBERRY PI 5 REQUIREMENT

- Raspberry Pi 5 Board
 - 2GB RAM (minimum)
 - Processor: Quad-core 64 bit Arm Cortex-A76 CPU
 - Potential [source 1](#) (Recommended, direct from source)
 - Potential [source 2](#) (verified seller, go for board only option)
- Case & Heat Sink
 - Completely optional (Mostly no need)
- Complete Package (**Hassel-free solution**)
 - Pi 5 2GB [Essentials Kit – Clear](#)
 - Pi 5 2 GB Kit [with SanDisk](#)



RASPBERRY PI 5 REQUIREMENT

- Memory Card (32 GB minimum)
 - Micro SD Card
 - Need to install OS (will work as a hard drive)
 - Potential source:
 - [Source 1](#)
 - [Source 2](#)
- SD/MicroSD Card Reader
 - Potential [Source 1](#)
 - [Source 2](#)



RASPBERRY PI 5 REQUIREMENT

- Power Source
 - 45W power adapter charging source
 - Potential [source 1](#), [source 2](#)
- Overall Recommendation For Raspberry PI 5 setup
 - Calculate the total of the mandatory items for Raspberry PI 5 setup, then check for a complete package (starter kit). Go for the budget friendly option.
 - As a first-time buyer, I recommend go for complete package (links provided in previous slides)



RASPBERRY PI 5 (BEST VALUE OPTION)



CanaKit Raspberry Pi 5 Essentials Kit - Turbine Clear

CanaKit

Pi 5 1GB Essentials Kit - Clear
PI5-1GB-STR32-C4-CLR-AC

CA\$ 139.95

[Add to Cart](#)

Pi 5 2GB Essentials Kit - Clear
PI5-2GB-STR32-C4-CLR-AC

CA\$ 169.95

[Add to Cart](#)

Pi 5 4GB Essentials Kit - Clear
PI5-4GB-STR32-C4-CLR-AC

CA\$ 229.95

[Add to Cart](#)

Pi 5 8GB Essentials Kit - Clear
PI5-8GB-STR32-C4-CLR-AC

CA\$ 319.95

[Add to Cart](#)

Pi 5 16GB Essentials Kit - Clear
PI5-16GB-STR32-C4-CLR-AC

CA\$ 494.95

[Add to Cart](#)

Features

RASPBERRY PI 5 (HIGH END, NOT REQUIRED)

Electronics › Computers & Accessories › Computer Components



[Click to see full view](#)

CanaKit Raspberry Pi 5 Starter Kit PRO - Turbine Black (128GB Edition) (8GB RAM)

Brand: CanaKit

4.6 ★★★★★ (1,384)

Amazon's Choice for "raspberry pi 5 amazon"

50+ bought in past month

\$349⁹⁹

Size: **8GB RAM**

Brand	CanaKit
Model name	Raspberry Pi 5
RAM memory installed size	8 GB
Memory storage capacity	128 GB
CPU model	MediaTek MT8125
CPU speed	2.4 GHz

▼ [See more](#)

REQUIRED HARDWARE

1. Electronics (Mandatory)

- Temperature/Humidity Sensor (DHT 11)
- LED's (Multi-color, {Red, Green, Yellow} mandatory),
- Buttons (3 ~ 6)
- Relays (5V)
- Motion Sensor (Passive Infrared (PIR)) - HC-SR501 PIR Sensor
- Ultrasonic Sensor - HC-SR04
- Resistors (10 Ω ~ 10 K Ω)
- Wires
- Breadboard (at least one)

15 Types of Sensor in IOT



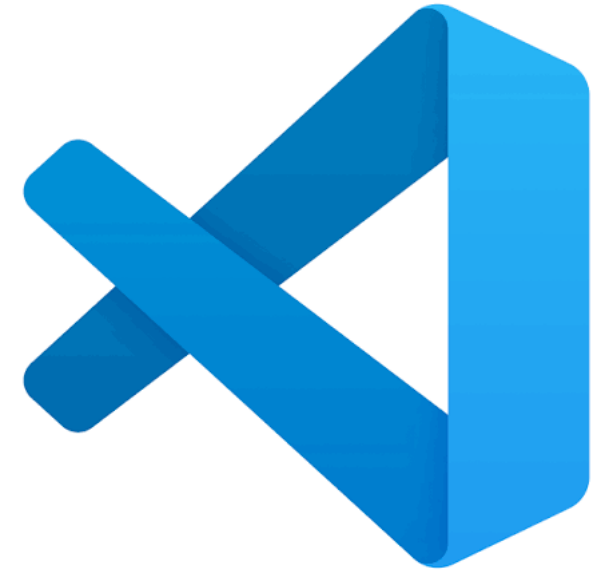
REQUIRED HARDWARE

1. Electronics (Complete Package – Much cheaper)

- Variation 1 : sensors, resistors, no LEDs
- Variation 2: everything
- **Recommendation**: There are lot of packages available with cheaper or higher price. When selecting a package, please make sure the mandatory electronic elements mentioned in previous slide are present in the package.
- You can check AliExpress as well for much cheaper option, but no validity of sensors working. It happened before.

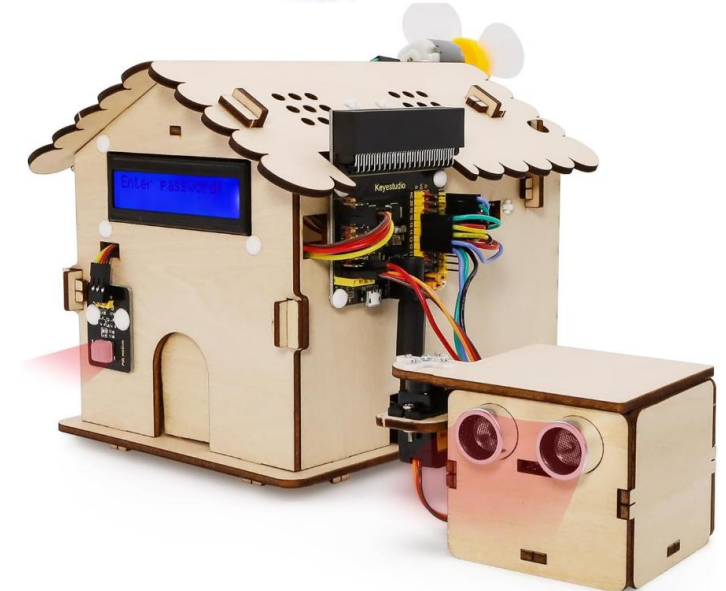
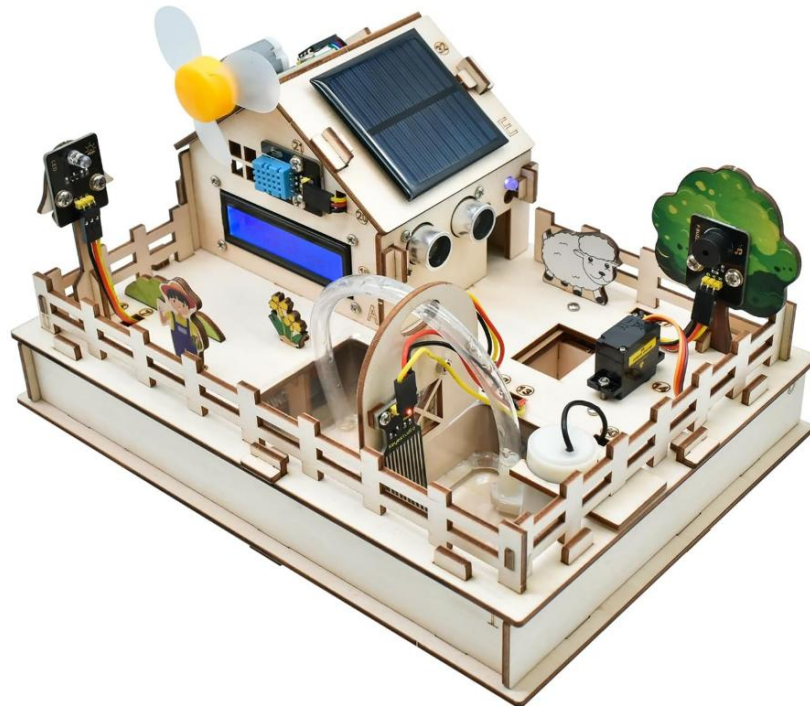
MEDIAGRAPHY/SOFTWARE

- [Visual Studio Code](#) (commonly known as VS Code) is a free, lightweight, and highly customizable source code editor developed by Microsoft.
- It is currently the most popular development environment in the world, supporting Windows, macOS, Linux, and web browsers. [[1](#), [2](#), [3](#), [4](#), [5](#)]



FINAL PROJECT OPTION

1. Line Tracking Robot
2. Smart House Monitoring System
3. Smart Farming



FINAL PROJECT SOURCE

1. Line Tracking Robot
 1. Potential Source 1
 2. Potential Source 2
2. Smart House Monitoring System
 1. Potential Source 1
3. Smart Farming
 1. Potential Source 1

FINAL PROJECT

■ Overall Heads-up & Recommendation

1. If you select either Smart home or Smart farming as your final project, you don't need to buy the sensor kits. The smart home and smart farming kits come with almost all the sensors we need for our class. You only need to buy the missing ones but listed as mandatory in the previous slides
2. If you select Smart Car as your final project, you don't need to build a website and control the car. You will be allowed to control the car by sending commands directly from your computer. But you need to buy the sensors or sensor kit. (Only two groups are allowed to do this project.)
3. The project description for each project will be provided later.

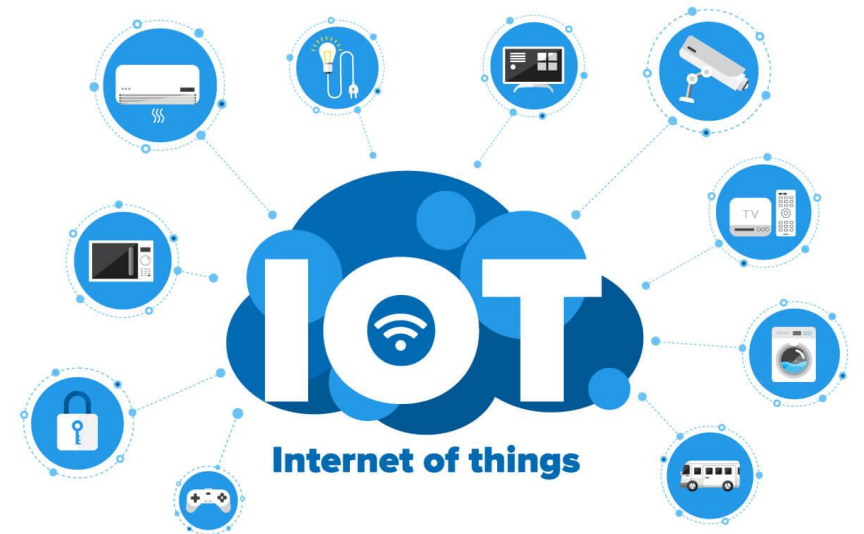


INTERNET OF THINGS



WHAT IS INTERNET-OF-THINGS (IOT)

- The Internet of Things (IoT) refers to the network of interconnected physical devices, vehicles, buildings, and other objects that are embedded with sensors, software, and other technologies, allowing them to collect and exchange data over the internet.
- The concept behind IoT is to enable these devices to communicate and interact with each other and with central systems, often without requiring human intervention.



TOP INTERNET-OF-THINGS (IOT) EXAMPLES TO KNOW

- Connected cars.
- Smart appliances.
- Connected security systems.
- Smart agriculture equipment.
- Connected retail.
- Connected healthcare monitors.
- Connected manufacturing equipment.
- Connected cities.
- Watch:

<https://www.youtube.com/watch?v=7yYb2224yH0>



IOT PROJECTS USING ARDUINO & RASPBERRY PI

- IoT Pollution monitoring
- IoT Color based sorting (garbage sorting)
- IoT Weather Station monitoring
- IoT ICU patient monitoring
- IoT smart dust bin
- IoT car parking monitoring and automation

Watch:

<https://www.youtube.com/watch?v=ZOmlaimMaJc>

<https://www.youtube.com/watch?v=WINVetH4ieg>

https://www.youtube.com/watch?v=_AlcRoqS65E&t=10s